Research ethics and consent

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete fictional proposal and permission audit.

Keep purpose, conditions, supplied evidence, proposed repair and unperformed actions together.

Keep one habit: Recall → W1–W3/W5 → independent opt-in scenario → research project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

Dataset and benchmark limitations

My remembered idea, example and correction

Ablation studies

My remembered idea, example and correction

Effect size and practical importance

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- This lesson audits fictional proposals under a supplied classroom policy; it performs no recruitment or real data collection. The policy calls for responsible review, plain-language purpose, affirmative opt-in and withdrawal handling before an actual collection. Submission for grading does not establish permission to copy essays into a study. Use fictional substitutes in this course.

  A proposed form or remedy does not prove review or consent occurred. Separate fictional adult drawing agreement covers one named pattern-recognition investigation and excludes public posting of originals and unrelated face-recognition research. A new request does not authorize excluded uses. Even a named use still needs its other review and handling conditions.

  The picture describes anonymous drawings without a traceable withdrawal mechanism; do not invent an ability to find every unlinked contribution. Explain declared withdrawal limits before participation. The adult example does not approve research with children or determine real institutional/legal requirements. Keep purpose, voluntary choice, burden, readers and handling visible in the paper audit.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Proposal automatically copies student essays into a study because they were submitted for grading. Show the intermediate steps and give the conclusion with its scope.

Main proposal step	Evidence supplied	Audit result under this policy
Student submits essay	Submission is for grading	Supports the grading purpose only
Team proposes automatic study copy	No separate research opt-in or review supplied	Research permission is not established
Assume submission implies agreement	Purpose changes without its required process	Unsupported inference
Repair for this course	Use fictional essay substitutes	No real records need to be copied

Main proposal step	Evidence supplied	Audit result under this policy
Any actual future study	Required review and voluntary permission before collection	Not performed or approved by this exercise

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  A proposed form or remedy does not prove review or consent occurred. Separate fictional adult drawing agreement covers one named pattern-recognition investigation and excludes public posting of originals and unrelated face-recognition research. A new request does not authorize excluded uses. Even a named use still needs its other review and handling conditions.

  The picture describes anonymous drawings without a traceable withdrawal mechanism; do not invent an ability to find every unlinked contribution. Explain declared withdrawal limits before participation. The adult example does not approve research with children or determine real institutional/legal requirements. Keep purpose, voluntary choice, burden, readers and handling visible in the paper audit.

- A report omits this qualification: This classroom policy exercise does not determine real legal or institutional requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- This lesson audits fictional proposals under a supplied classroom policy; it performs no recruitment or real data collection. The policy calls for responsible review, plain-language purpose, affirmative opt-in and withdrawal handling before an actual collection. Submission for grading does not establish permission to copy essays into a study. Use fictional substitutes in this course.

  A proposed form or remedy does not prove review or consent occurred. Separate fictional adult drawing agreement covers one named pattern-recognition investigation and excludes public posting of originals and unrelated face-recognition research. A new request does not authorize excluded uses. Even a named use still needs its other review and handling conditions.

  The picture describes anonymous drawings without a traceable withdrawal mechanism; do not invent an ability to find every unlinked contribution. Explain declared withdrawal limits before participation. The adult example does not approve research with children or determine real institutional/legal requirements. Keep purpose, voluntary choice, burden, readers and handling visible in the paper audit.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled research ethics and **consent evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of research ethics and **consent**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Research ethics and **consent**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Research ethics and **consent**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

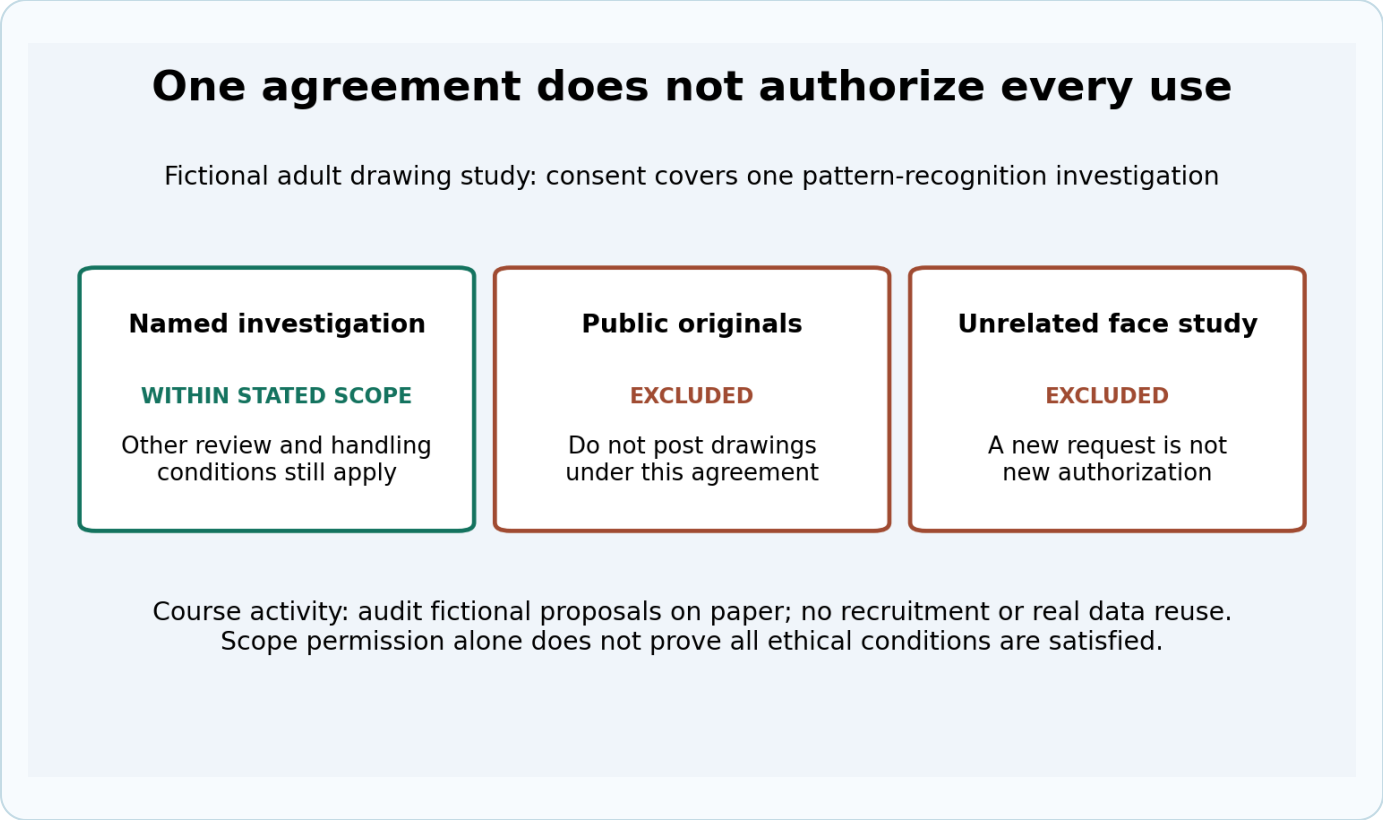
> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Consent** is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.
- **Consent** covers this study only. It excludes public posting of originals and unrelated face-recognition research.
- Identify which uses the drawing **consent** permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/19

Research ethics and consent

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State the purpose

Explain the drawing task and planned use before participation.



2. Respect exclusions

Do not publish originals or transfer them to the unrelated project.



3. Keep the agreement

Record permitted uses and the approved withdrawal procedure.

CHANGE ONE THING

A new project requests the same drawings for an excluded purpose.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- **Consent** is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.
- Which labelled step matches this detail? Record permitted uses and the approved withdrawal procedure.
- 1. State the purpose
- 2. Keep the agreement
- 3. Respect exclusions
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep the agreement Record permitted uses and the approved withdrawal procedure.

Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Changed-case question and optional hint

- Changed-case practice: A new project requests the same drawings for an excluded purpose. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Do not reuse them under the existing **consent**; seek the required new authorisation.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Do not reuse them under the existing **consent**; seek the required new authorisation. The new purpose falls outside the stated agreement.

Repair and retain the record

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Research ethics and consent: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

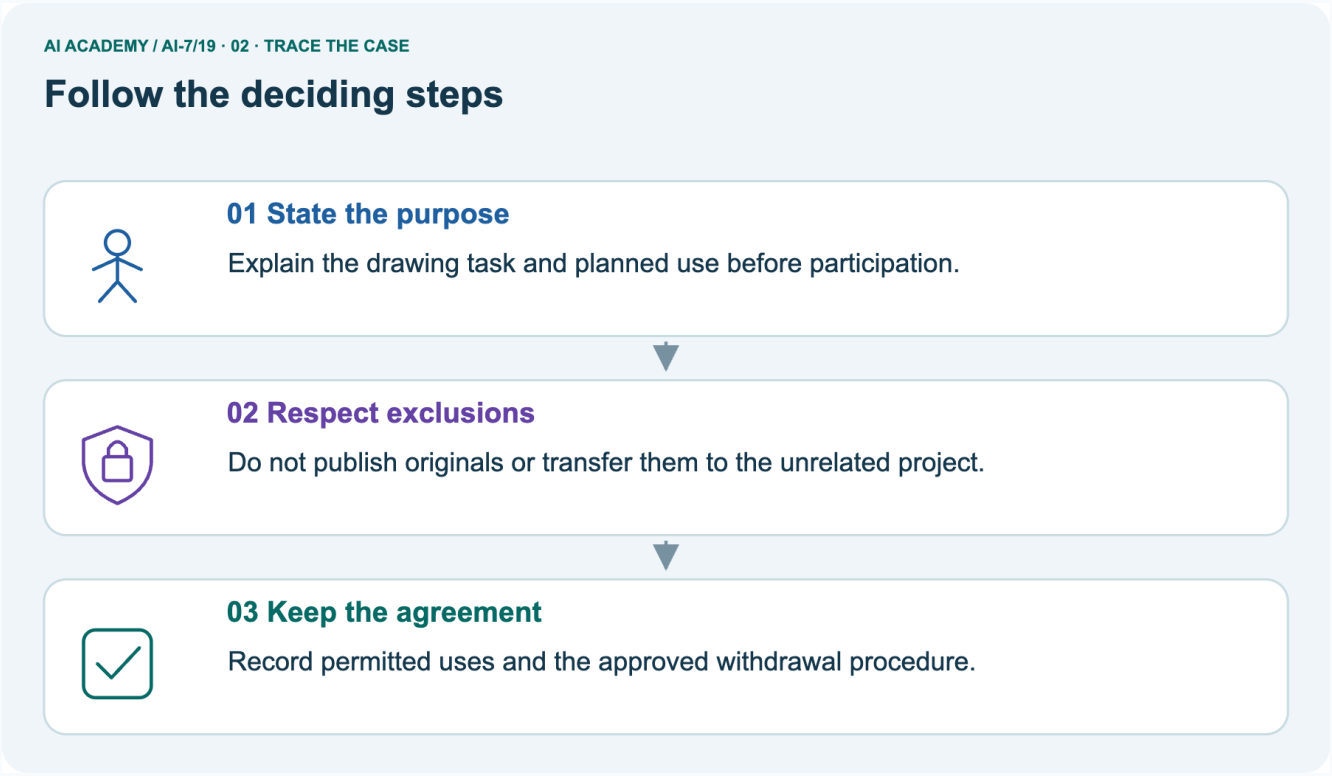
Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation. Supplied facts: **Consent** covers this study only.

It excludes public posting of originals and unrelated face-recognition research.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State the purpose: Explain the drawing task and planned use before participation. → Respect exclusions: Do not publish originals or transfer them to the unrelated project. → Keep the agreement: Record permitted uses and the approved withdrawal procedure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Research ethics and consent — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Research ethics and **consent** — **evidence** trace. Proposal automatically copies student essays into a study because they were submitted for grading. | Use fictional records for this course.

For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in **consent** and withdrawal handling before collection.

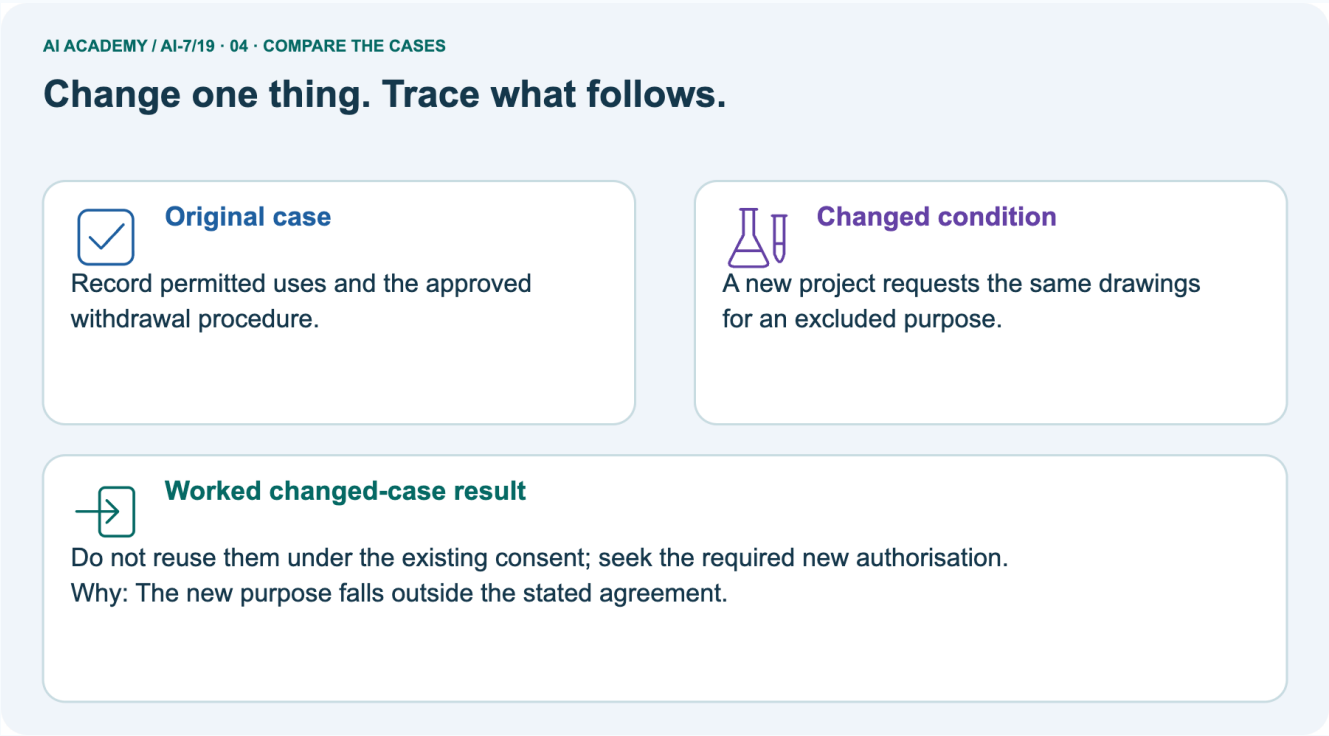
Do not implement real recruitment in this exercise. | The grading purpose does not establish research permission under the policy.

Redesign around fictional data or obtain required review and voluntary permission before collection.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

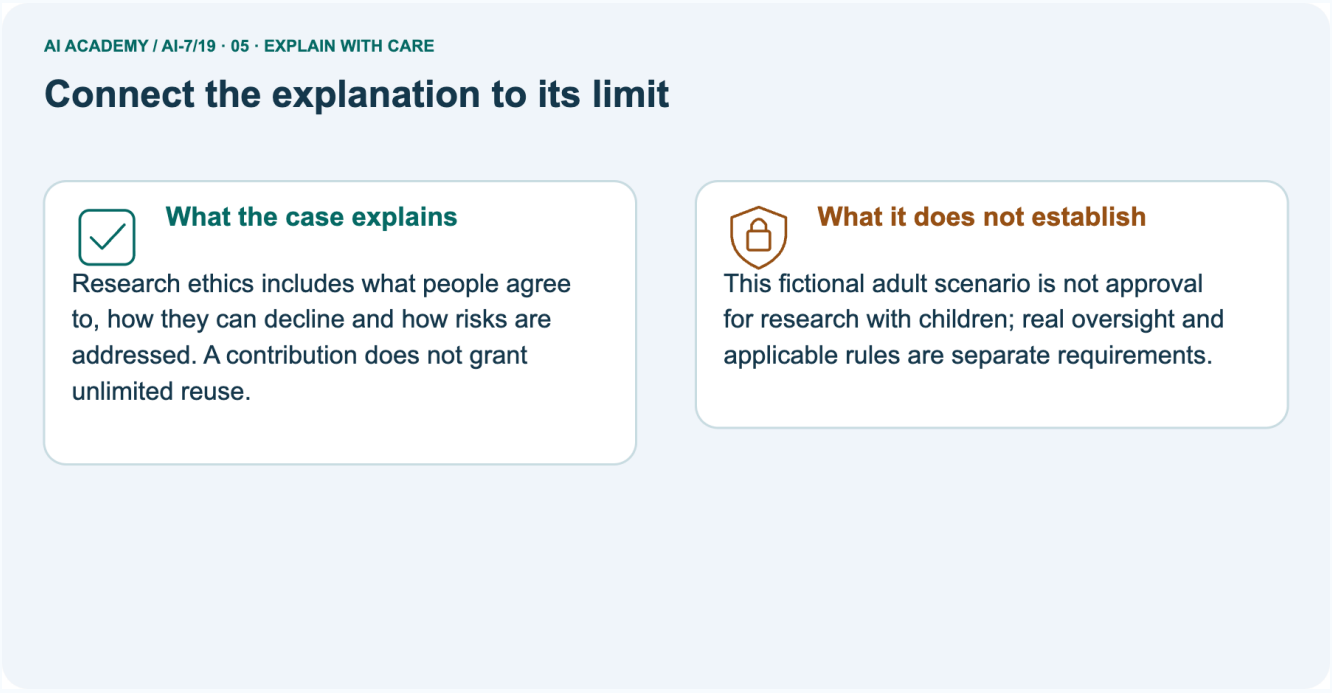
Original case: Record permitted uses and the approved withdrawal procedure. Changed condition: A new project requests the same drawings for an excluded purpose.

Expected result: Do not reuse them under the existing **consent**; seek the required new authorisation. Reason: The new purpose falls outside the stated agreement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

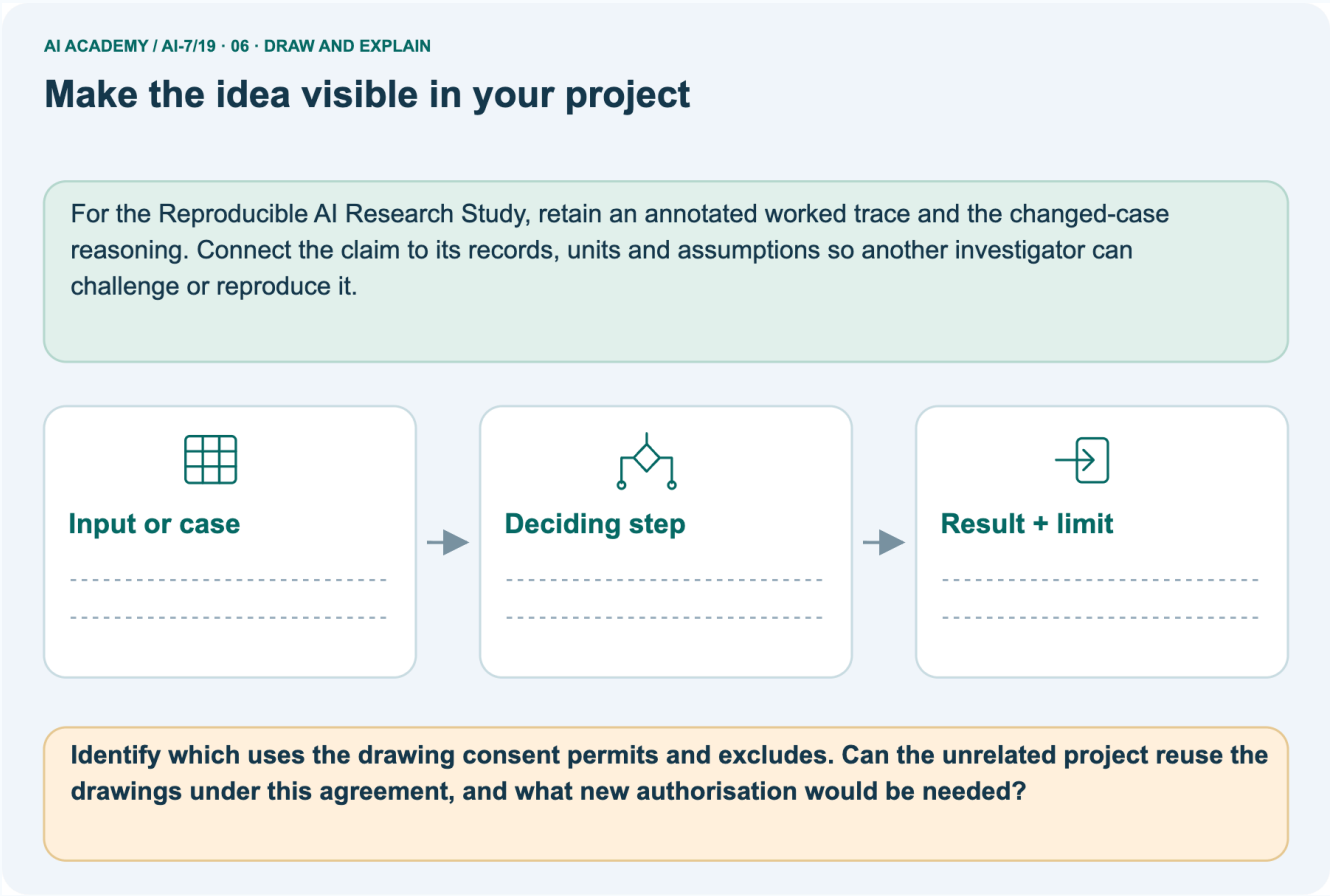
Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Limit: This fictional adult scenario is not approval for research with children; real oversight and applicable rules are separate requirements.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Identify which uses the drawing **consent** permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Research ethics and consent: the case

02. Follow the deciding steps

03. Research ethics and consent — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.